Sangjun (Samuel) Ko

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Work Experience

Kimberly-Clark — North America Family Care R&D Neenah, Wisconsin
Statistician (contract assignment through Staffmark Group) June 2025 – Present

- Rebuilt Kimberly-Clark's core diaper-leakage modeling engine from legacy SAS into a containerized Python/Flask web app; resolved whole-plot vs. sub-plot error structure partitioning in split-plot repeated-measures trials, achieving benchmark parity across 9 historical studies and reducing report generation time from days to minutes for 50+ researchers.
- Designed balanced factorial equivalence studies ($N = 96$) to qualify replacement abrasive mandrels across two instrument lines; detected a persistent 0.7–1.3 mg negative measurement bias that disproved vendor equivalence, halting premature line adoption and driving supplier re-machining.
- Designed multi-factor screening DOEs (including 96-run mechanical cart comparisons and 48-run fluid insult rate studies) to correlate accelerated benchtop testing with 3-month consumer home-use test (HUT) data, isolating key design drivers and screening prototypes in weeks.
- Engineered an automated time-series data reduction pipeline in Python for transient sensor curves; modeled moisture dissipation dynamics to identify a 15–17 minute plateau, establishing an empirical 14–15 minute reading rule that eliminated operator timing variation.
- Deployed six production web modules on Kimberly-Clark's internal cloud computing hub, operationalizing multivariate product-performance models (19 laboratory predictors → 8 consumer attributes) and automated Gage R&R sample-size calculators into self-service tools for R&D.

Applied Technologies for Learning in Arts & Sciences (ATLAS) Champaign, Illinois
AI/ML Intern Team Leader January 2024 – December 2024

- Led five interns and coordinated deliverables with three managers while developing and evaluating Python machine-learning and NLP models (BERT, RoBERTa, XGBoost, Random Forests) using datasets containing 100,000+ records.

University of Illinois Urbana-Champaign Champaign, Illinois
Undergraduate Course Assistant — Data Science and Professional Statistics August 2023 – May 2025

- Mentored 300+ students in foundational data science and statistical consulting; managed Git-based instructional content for a learning platform serving 1,000+ students across STAT 107 and STAT 443.

Selected Project

Graduation Prediction Model September 2024 – December 2024

- Increased model AUC from 0.65 to 0.82 through feature engineering and outlier review across 50,000+ student records; built an interactive Streamlit interface using 12 key predictors to deliver real-time predictions for university administrators.

Education

University of Illinois Urbana-Champaign Champaign, Illinois
Bachelor of Science in Statistics, Mathematics Minor; GPA: 3.93/4.00 January 2023 – May 2025

University of Michigan–Shanghai Jiao Tong University Joint Institute Shanghai, China
Coursework toward a B.E. in Mechanical Engineering; transferred to UIUC; First Class Scholarship September 2019 – December 2020

Technical Skills

Statistical Methods:	Design of Experiments (DOE), Test Method Validation (TMV), Gauge R&R/MSA, mixed models, split-plot ANOVA, tolerance intervals, attribute agreement, bias/precision analysis
Programming & Web:	Python, R, SQL, JavaScript, HTML/CSS, Flask, Streamlit, Docker, Git, Excel/VBA
Libraries & Tools:	Pandas, NumPy, SciPy, statsmodels, scikit-learn, TensorFlow, JMP, SAS, Tableau
Languages:	Korean and Spanish (native); English (fluent)

Service

Volunteer, Appleton Public Library August 2025 – Present

- Tutor middle- and high-school students in math weekly; previously supported "Game On" teen programs.